Evolutionary Computation

Constraint Handling in Evolutionary Algorithms (I)

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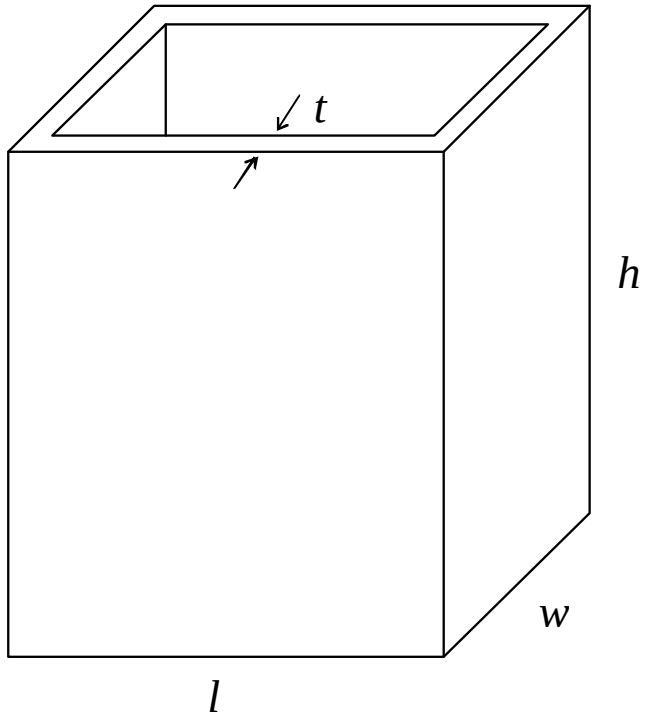
Outline of Topics

1. Motivating examples
   - Engineering Optimisation Problems
2. Constrained Optimisation
3. Constrained Handling Techniques in EAs
   - Penalty Function

Example 1: Cubic Vessel Design

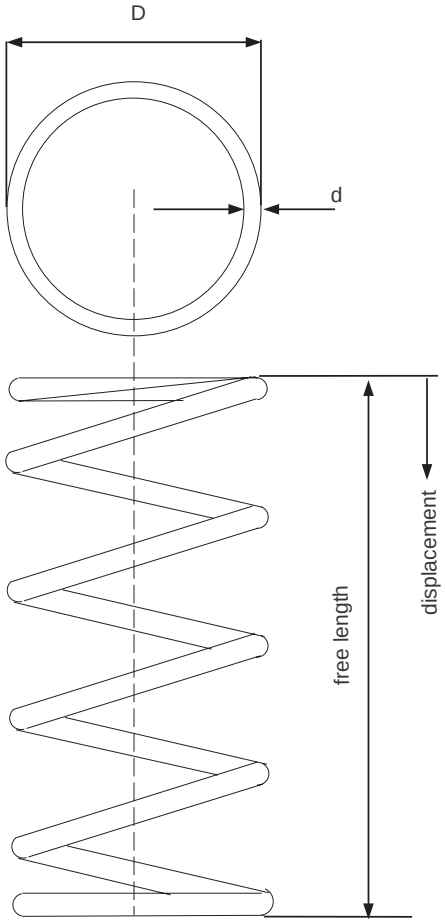
- **Aims**: to find an optimal values of length l, width w, high h, and thickness t, to minimize the material consumption (or equivalently, the surface area).
- **Constraints**: material consumption cannot be minimise infinitely since there are some requirements:
  - Restrictions imposed by government and corporate regulations, e.g., shapes or the maximum capacity
  - Quality requirements, e.g., the maximum deflection should be less than an allowable value.



Example 2: Spring design

- **Aims** to minimize the weight of a tension/compression spring. All design variables are continuous (See my paper).
- **Four constraints**:
  - Minimum deflection
  - Shear stress
  - Surge frequency
  - Diameters
- Let the wire diameter $d = x_1$, the mean coil diameter $D = x_2$ and the number of active coils $N = x_3$
- There are boundaries of design variables:

$$0.05 \leq x_1 \leq 2,\ 0.25 \leq x_2 \leq 1.3,\ 2 \leq x_3 \leq 15$$



Constrained Optimisation Example: Spring design

- Minimize

$$f(X) = (x_3 + 2)x_2 x_1^2 \tag{1}$$

subject to:

$$g_1(X) = 1 - \frac{x_2^3 x_3}{71785 x_1^4} \leq 0 \tag{2}$$

$$g_2(X) = \frac{4x_2^2 - x_1 x_2}{12566(x_2 x_1^3 - x_1^4)} + \frac{1}{5108 x_1^2} - 1 \leq 0 \tag{3}$$

$$g_3(X) = 1 - \frac{140.45 x_1}{x_2^2 x_3} \leq 0 \tag{4}$$

$$g_4(X) = \frac{x_2 + x_1}{1.5} - 1 \leq 0 \tag{5}$$

Engineering Optimisation

- **Engineering optimisation (Design Optimisation):** to find the best combination of design variables that optimises designer's preference (design objective) and satisfies certain requirements (constraints).
- **Design variables**: A design variable is under the control of designer and could have an impact on the solution of the optimization problem
- Types of design variables can be:
  - Continuous
  - Integer (including binary)
  - Set of variables: designers are required to choose the design variables from a list of recommended values from design standards
- **Design objective**: represents the desires of the designers, e.g., to maximize profit or minimize cost.

Engineering Optimisation

- **Constraints:** Designer's desires cannot be optimized infinitely because of
  - Limited resources: e.g., budget or materials that can be used in product development.
  - Other restrictions:
- A design constraint is "rigid" since usually it needs to be satisfied strictly
- Engineering optimisation, as well many real-world optimisation problems are **constrained optimisation problems**

Engineering Optimisation Software

- Optimisation software list
- Matlab Optimisation Toolbox
- Matlab Global Optimisation Toolbox
- Red Cedar Technology
- Omniquest: see
  - "How to create an algorithm that will perform the design process more efficiently and reliably?" - Omniquest founder, Dr. Vanderplaats' interview
  - Multi-start Optimization

What Is Constrained Optimisation?

- The general constrained optimisation problem:

$$\min_{\mathbf{x}}\{f(\mathbf{x})\}$$

subject to

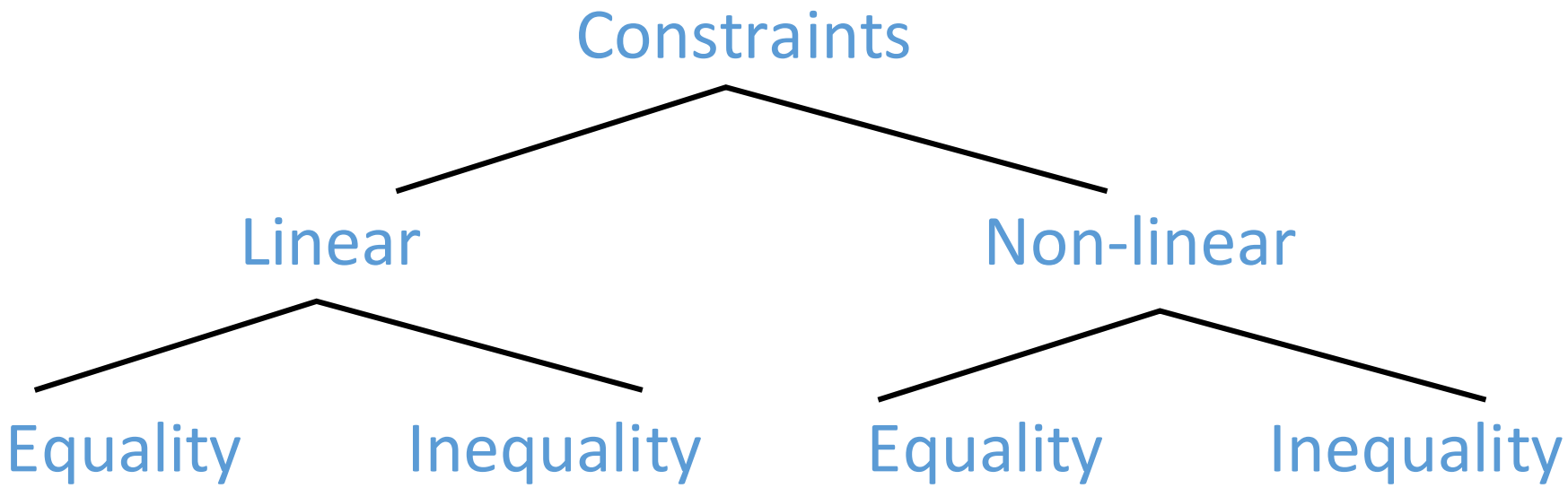
$$g_i(\mathbf{x}) \leq 0,\ i = 1, \cdots, m$$

$$h_j(\mathbf{x}) = 0,\ j = 1, \cdots, p$$

where $\mathbf{x}$ is the n dimensional vector, $x = (x_1, x_2, \cdots, x_n)$; $f(\mathbf{x})$ is the objective function; $g_i(\mathbf{x})$ is the inequality constraint; and $h_j(\mathbf{x})$ is the equality constraint.

- Denote the whole search space as $\mathcal{S}$ and the feasible space as $\mathcal{F}$, $\mathcal{F} \in \mathcal{S}$.
- Note: the global optimum in $\mathcal{F}$ might not be the same as that in $\mathcal{S}$.

Different Types of Constraints



- Linear constraints are relatively easy to deal with
- Non-linear constraints can be hard to deal with

Constraint Handling Techniques in Evolutionary Algorithms

- **The purist approach (dead penality)**: rejects all infeasible solutions in search
  - **Question**: What is the main drawback of the purist approach?
- **The penalty function approach**: converts a constrained problem into an unconstrained one by introducing a penalty function into the objective function.
- **The repair approach**: maps (repairs) an infeasible solution into a feasible one.
- **The hybrid approach** mixes two or more different constraint handling techniques.

Penalty Function Approach: Introduction

- New Objective Function = Original Objective Function + Penalty Coefficient * Degree Of Constraint Violation (note: minimisation)
- The general form of the penalty function approach:

$$f'(\mathbf{x}) = f(\mathbf{x}) + \left(\sum_{i=1}^{m} r_i G_i(\mathbf{x}) + \sum_{j=1}^{p} c_j H_j(\mathbf{x}) \right)$$

where $f'(\mathbf{x})$ is the new objective function to be minimised, $f(\mathbf{x})$ is the original objective function, $r_i \in \mathbb{R}^+$ and $c_j \in \mathbb{R}^+$ are penalty factors (coefficients), and $G_i \geq 0$ and $H_j \geq 0$ are penalty functions for inequality and equality constraints, respectively:

$$G_i(\mathbf{x}) = \max(0, g_i(\mathbf{x}))^\beta, \; H_j(\mathbf{x}) = \max(0, |h_j(\mathbf{x})|)^\gamma,$$

where β and γ are usually chosen as 2.

Penalty Function Approach: Techniques

- **Static Penalties**: The penalty function is pre-defined and fixed during evolution.
- **Dynamic Penalties**: The penalty function changes according to a pre-defined sequence, which often depends on the generation number.
- **Adaptive and Self-Adaptive Penalties**: The penalty function changes adaptively.

Static Penalty Functions

- Static penalty functions general form:

$$f'(\mathbf{x}) = f(\mathbf{x}) + \sum_{i=1}^{m} r_i (G_i(\mathbf{x}))^2$$

  where $r_i \in \mathbb{R}^+$ are predefined and fixed.
- Equality constraints can be converted into inequality ones:

$$h_j(\mathbf{x}) \Longrightarrow h_j(\mathbf{x}) - \epsilon \leq 0$$

  where $\epsilon > 0$ is a small number.
- Simple and easy to implement.
- Requires rich domain knowledge to set r_i.
- r_i $(i = 1, \cdots, m + p)$ can be divided into a number of different levels. When to use which is determined by a set of heuristic rules.

Dynamic Penalty Functions

- Dynamic Penalties general form:

$$f'(\mathbf{x}) = f(\mathbf{x}) + r(t)\sum_{i=1}^{m} G_i^2(\mathbf{x}) + c(t)\sum_{j=1}^{p} H_j^2(\mathbf{x})$$

  where $r(t)$ and $c(t)$ are two penalty coefficients.
- General principle: the large the generation number t, the larger the penalty coefficients $r(t)$ and $c(t)$.
- **Question:** why the larger the generation number, the larger the penalty coefficients?

Dynamic Penalty Functions

- Common dynamic penalty coefficients:
  - **Polynomials**: $r(t) = \sum_{k=1}^{N} a_{k-1} t^{k-1}$, $c(t) = \sum_{k=1}^{N} b_{k-1} t^{k-1}$ where a_k and b_k are user-defined parameters.
  - **Exponentials**: $r(t) = e^{at}$, $c(t) = e^{bt}$ where a and b are user-defined parameters.
  - **Hybrid**: $r(t) = e^{\sum_{k=1}^{N} b_{k-1} t^{k-1}}$, $c(t) = e^{\sum_{k=1}^{N} b_{k-1} t^{k-1}}$

Penalty function, Fitness Function and Selection

- Let static penalty function $\Phi(\mathbf{x}) = f(\mathbf{x}) + rG(\mathbf{x})$, where $G(\mathbf{x}) = \sum_{i=1}^{m} G_i(\mathbf{x})$, $G_i(\mathbf{x}) = \max\{0, g_i(\mathbf{x})\}$, and $r \in \mathbb{R}^+$
- **Question**: How does r affect an Evolutionary Algorithm? i.e., which parts?

Generic Evolutionary Algorithm

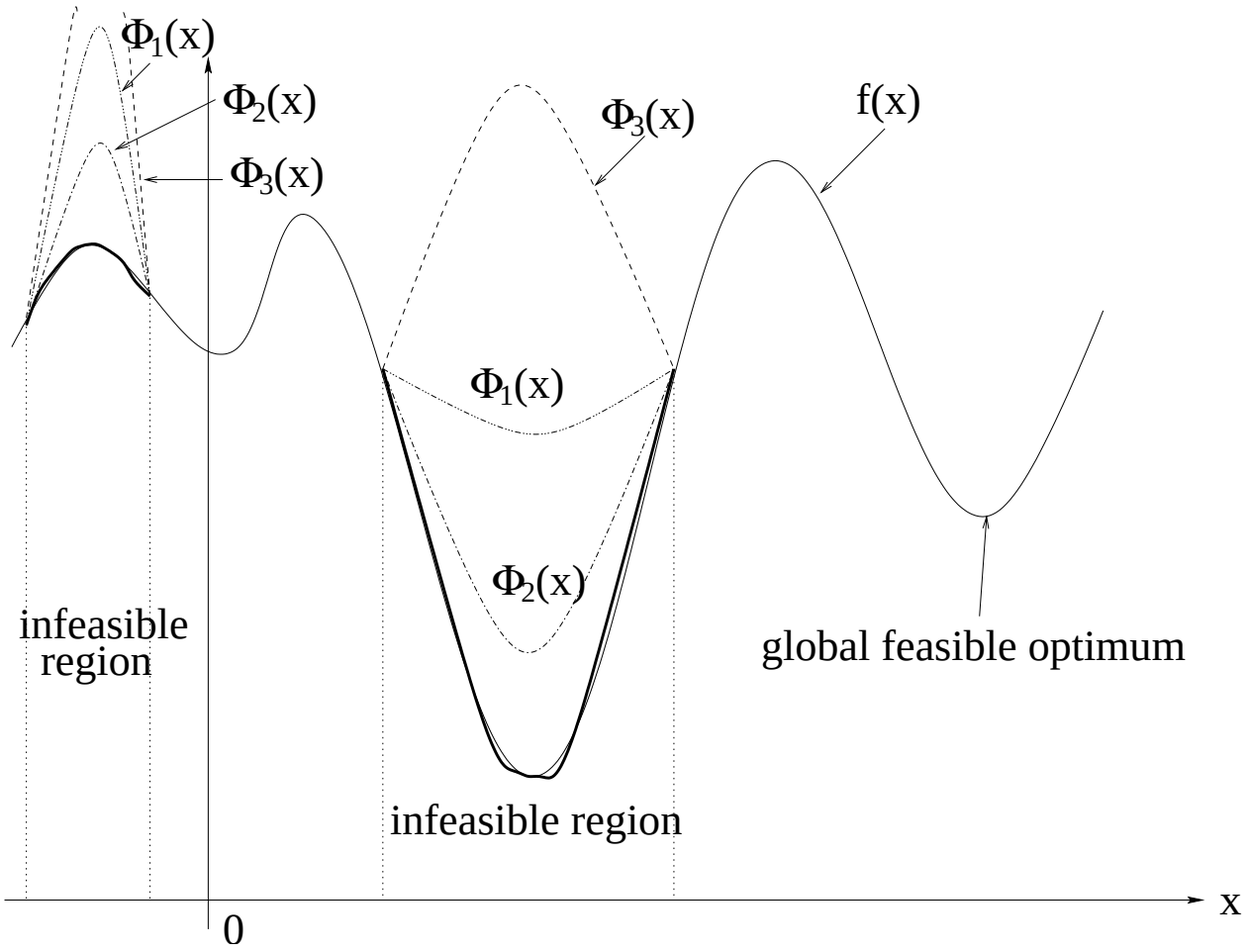
$\mathbf{X}_0$:= generate initial population of solutions
terminationflag := false
t := 0
Evaluate the fitness of each individual in $\mathbf{X}_0$.
while (terminationflag != true)
    Selection: Select parents from $\mathbf{X}_t$ based on their fitness.
    Variation: Breed new individuals by applying variation operators to parents
    Fitness calculation: Evaluate the fitness of new individuals.
    Reproduction: Generate population $\mathbf{X}_{t+1}$ by replacing least-fit individuals
    t := t + 1
    If a termination criterion is met: terminationflag := true
Output x_{best}

Penalty function, Fitness Function and Selection

- **Minimisation problem** with a penalty function $\Phi(\mathbf{x}) = f(\mathbf{x}) + rG(\mathbf{x})$
- Given two individual $\mathbf{x_1}$ and $\mathbf{x_2}$, their fitness values are now determined by $\Phi(\mathbf{x})$
- Because fitness values are used primarily in selection: Changing fitness $\Longleftrightarrow$ changing selection probabilities
- Selecting $\mathbf{x_1} \Rightarrow \Phi(\mathbf{x_1}) < \Phi(\mathbf{x_2} \Rightarrow f(\mathbf{x_1}) + rG(\mathbf{x_1}) < f(\mathbf{x_2}) + rG(\mathbf{x_2})$, rearrange it, we have $f(\mathbf{x_1}) - f(\mathbf{x_2}) + r[G(\mathbf{x_1}) - G(\mathbf{x_2})] < 0$. When this inequality does not hold, we select $\mathbf{x_2}$ (**Note**: $r \in \mathbb{R}^+$)
  - $f(\mathbf{x_1}) \leq f(\mathbf{x_2})$ and $G(\mathbf{x_1}) \leq G(\mathbf{x_2})$: r has no impact on the comparison
  - $f(\mathbf{x_1}) < f(\mathbf{x_2})$ and $G(\mathbf{x_1}) > G(\mathbf{x_2})$: Increasing r will eventually change the comparison
  - $f(\mathbf{x_1}) > f(\mathbf{x_2})$ and $G(\mathbf{x_1}) < G(\mathbf{x_2})$: Decreasing r will eventually change the comparison
- **Answer**: In essence, different $r \in \mathbb{R}^+$ lead to different ranking of individual in the population

Penalties and Fitness Landscape Transformation

- Different penalty functions lead to different new objective functions.
- **Question**: For the following minimisation problem, which penalty function should we avoid? Why?



Penalty Functions Demystified

- Penalty function essentially:
  - Transforms fitness
  - Changes rank $\rightarrow$ changes selection
- Why not change the rank directly in an EA?
- We will introduce Stochastic Ranking in our next lecture